CloudFront & Global Accelerator

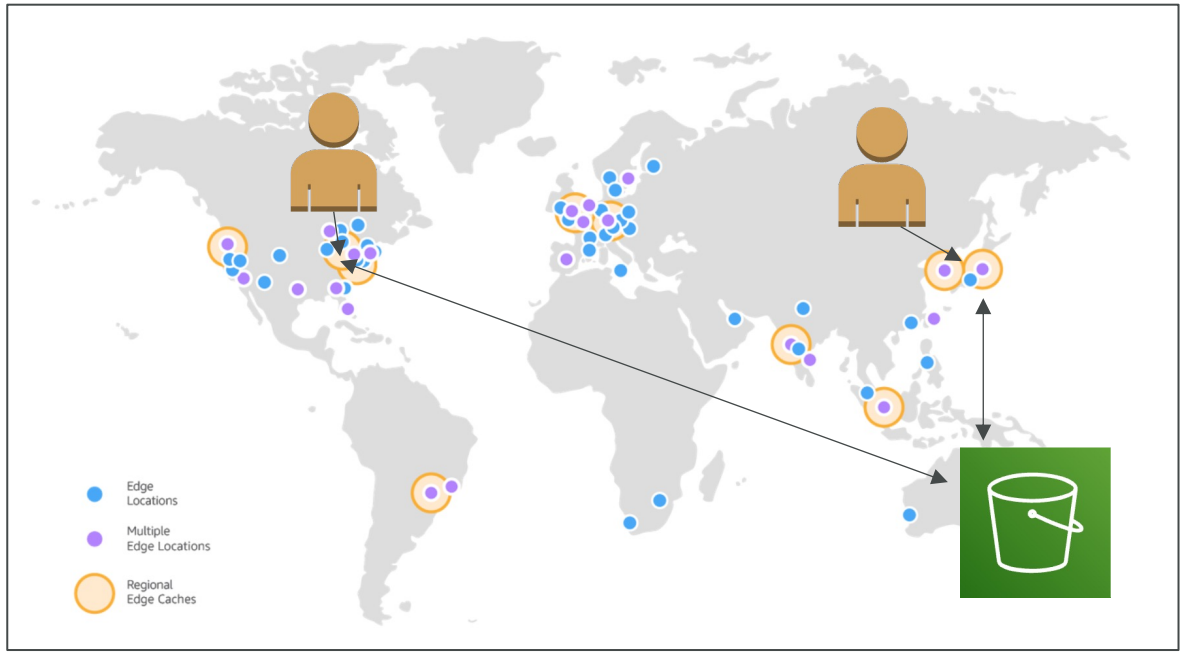
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Amazon CloudFront



- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- Improves users experience
- Hundreds of Points of Presence globally (edge locations, caches)
- DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall



Source: https://aws.amazon.com/cloudfront/features/?nc=sn&loc=2

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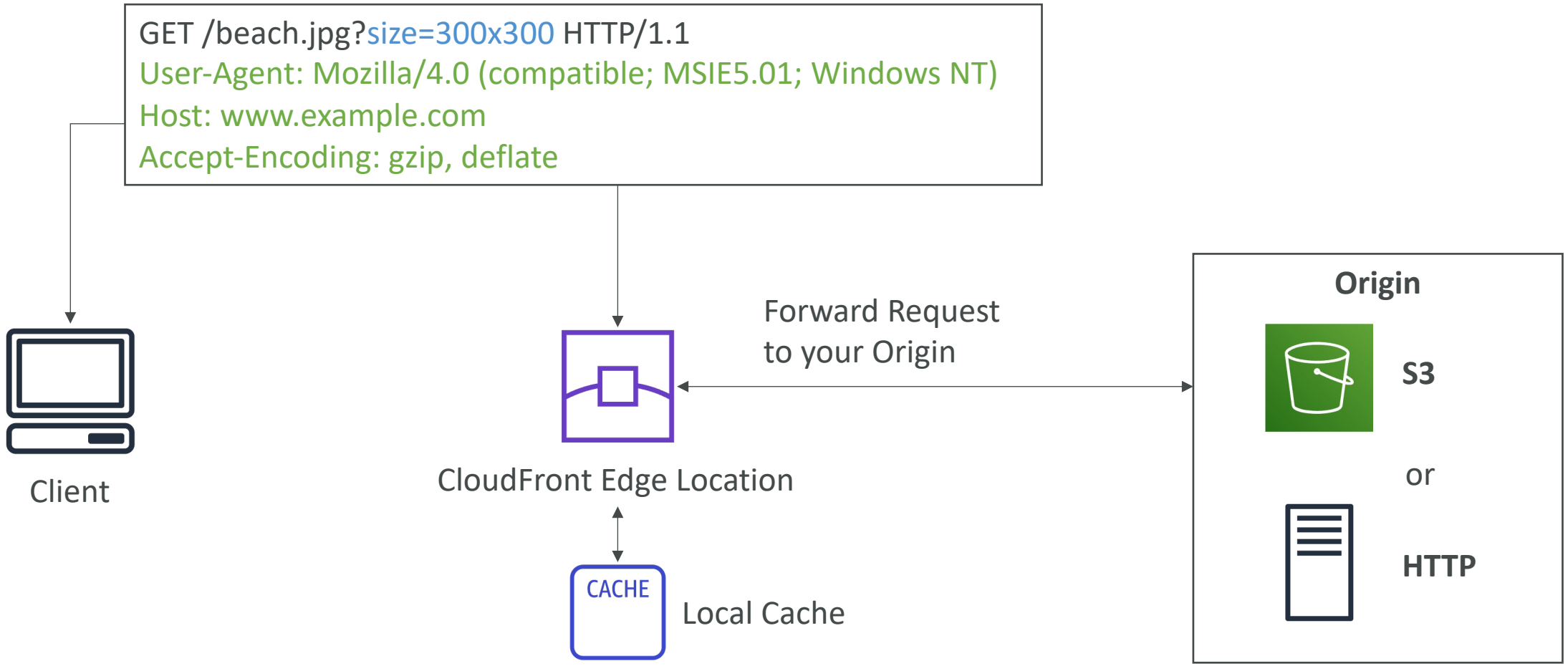
CloudFront – Origins

- S3 bucket
  - For distributing files and caching them at the edge
  - For uploading files to S3 through CloudFront
  - Secured using Origin Access Control (OAC)
- VPC Origin
  - For applications hosted in VPC private subnets
  - Private Application Load Balancer / Network Load Balancer / EC2 Instances
- Custom Origin (HTTP)
  - S3 website (must first enable the bucket as a static S3 website)
  - Any public HTTP backend you want (example: Public ALB)

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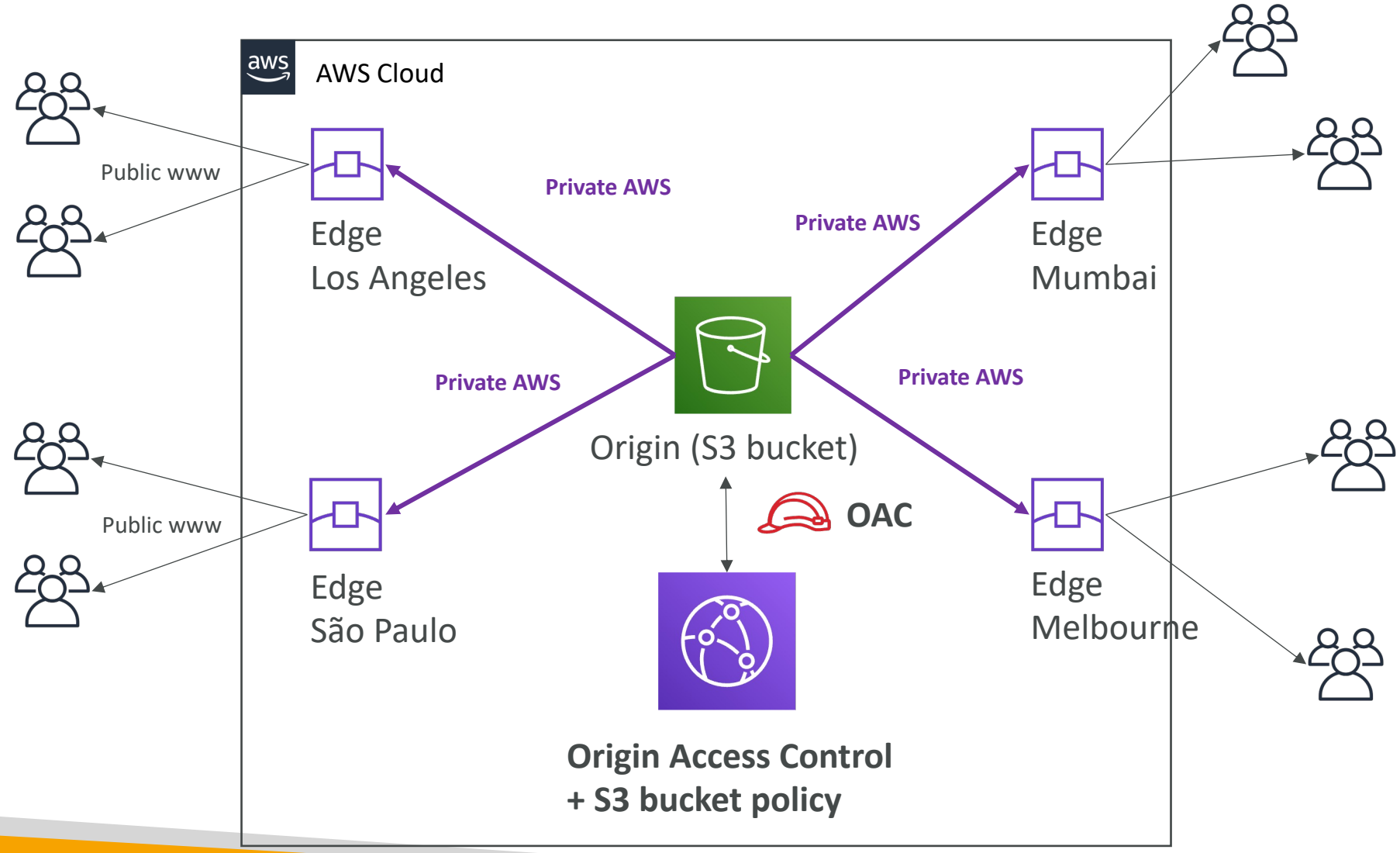
CloudFront at a high level



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CloudFront – S3 as an Origin



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CloudFront vs S3 Cross Region Replication

- CloudFront:
  - Global Edge network
  - Files are cached for a TTL (maybe a day)
  - **Great for static content that must be available everywhere**

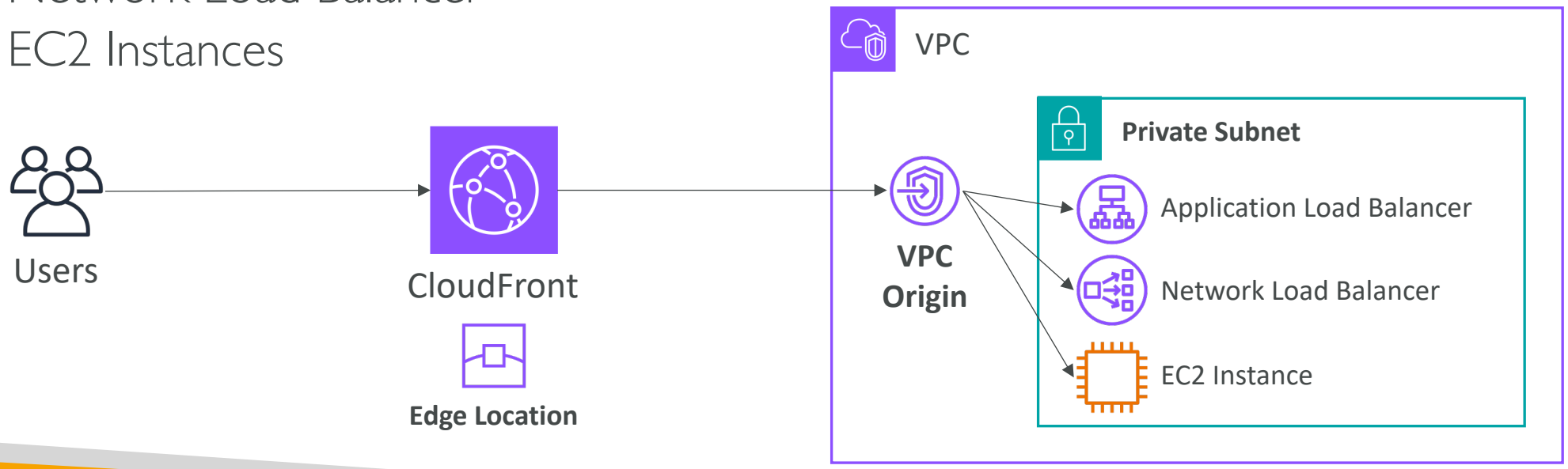
- S3 Cross Region Replication:
  - Must be setup for each region you want replication to happen
  - Files are updated in near real-time
  - Read only
  - **Great for dynamic content that needs to be available at low-latency in few regions**

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CloudFront – ALB or EC2 as an origin Using VPC Origins

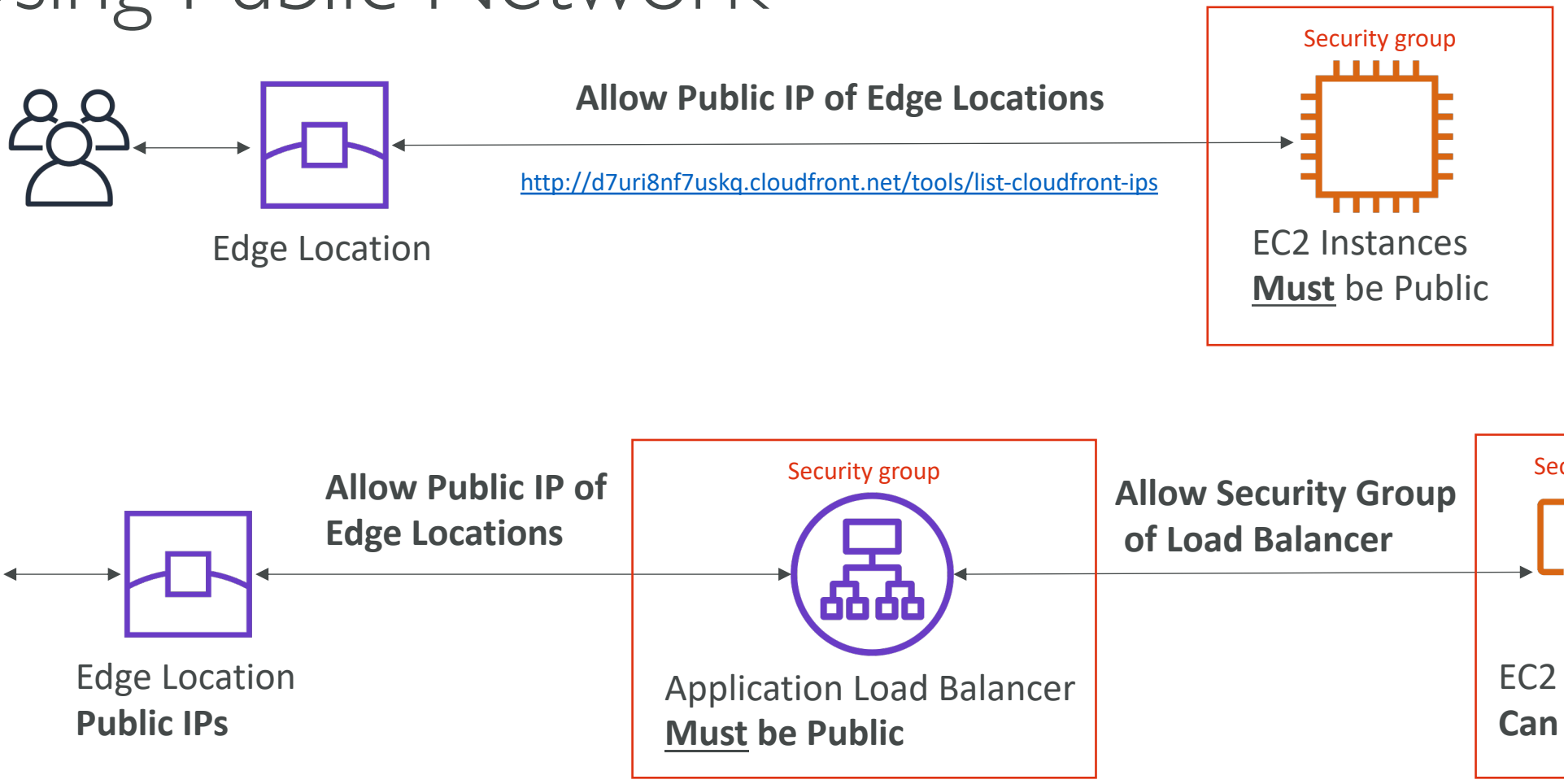
- Allows you to deliver content from your applications hosted in your VPC private subnets (no need to expose them on the Internet)
- Deliver traffic to **private**:
  - Application Load Balancer
  - Network Load Balancer
  - EC2 Instances



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CloudFront – ALB or EC2 as an origin Using Public Network



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CloudFront Geo Restriction

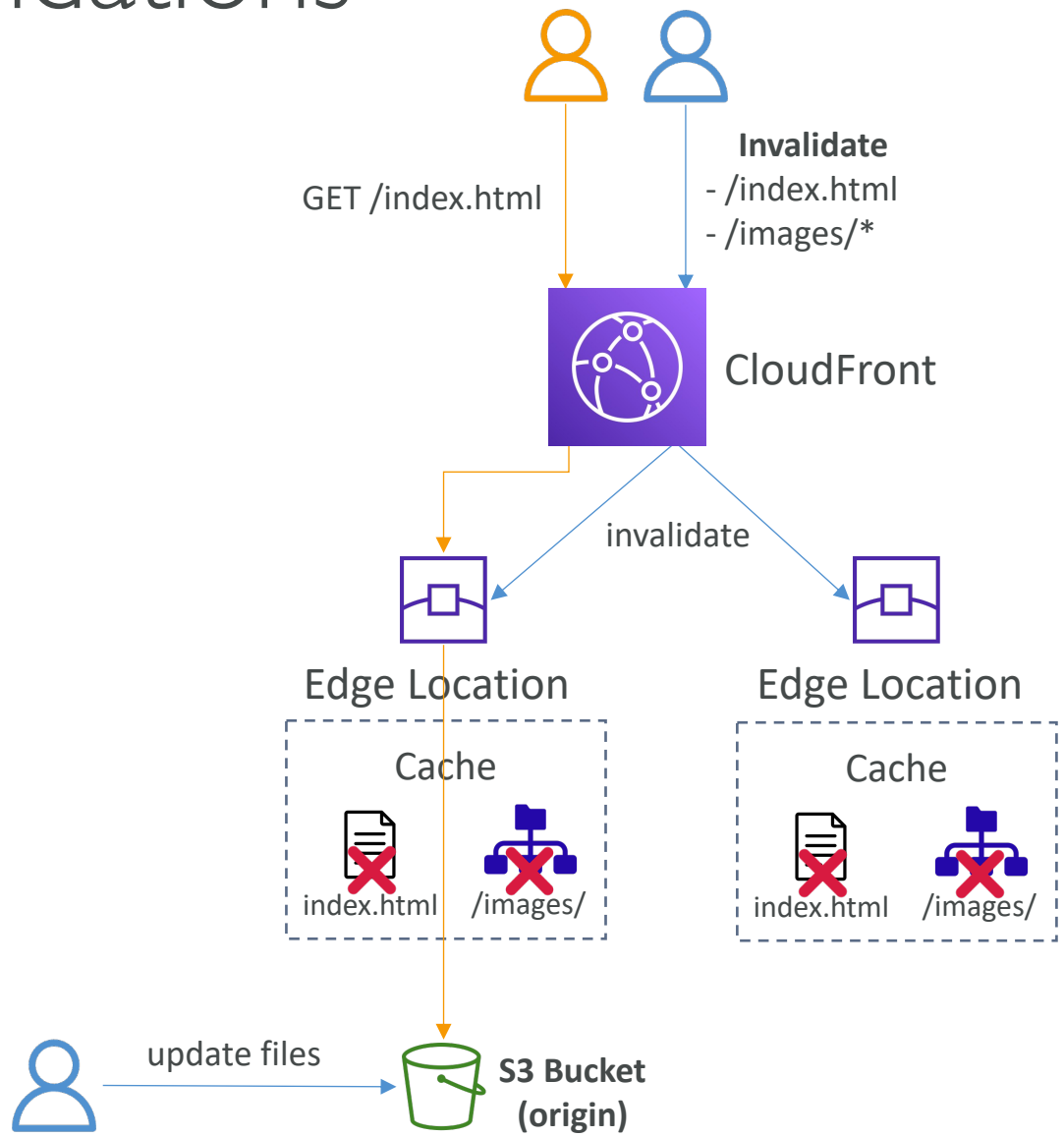
- You can restrict who can access your distribution
  - **Allowlist:** Allow your users to access your content only if they're in one of the countries on a list of approved countries.
  - **Blocklist:** Prevent your users from accessing your content if they're in one of the countries on a list of banned countries.
- The "country" is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content

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CloudFront – Cache Invalidations

- In case you update the back-end origin, CloudFront doesn't know about it and will only get the refreshed content after the TTL has expired
- However, you can force an entire or partial cache refresh (thus bypassing the TTL) by performing a **CloudFront Invalidation**
- You can invalidate all files (*) or a special path (/images/*)

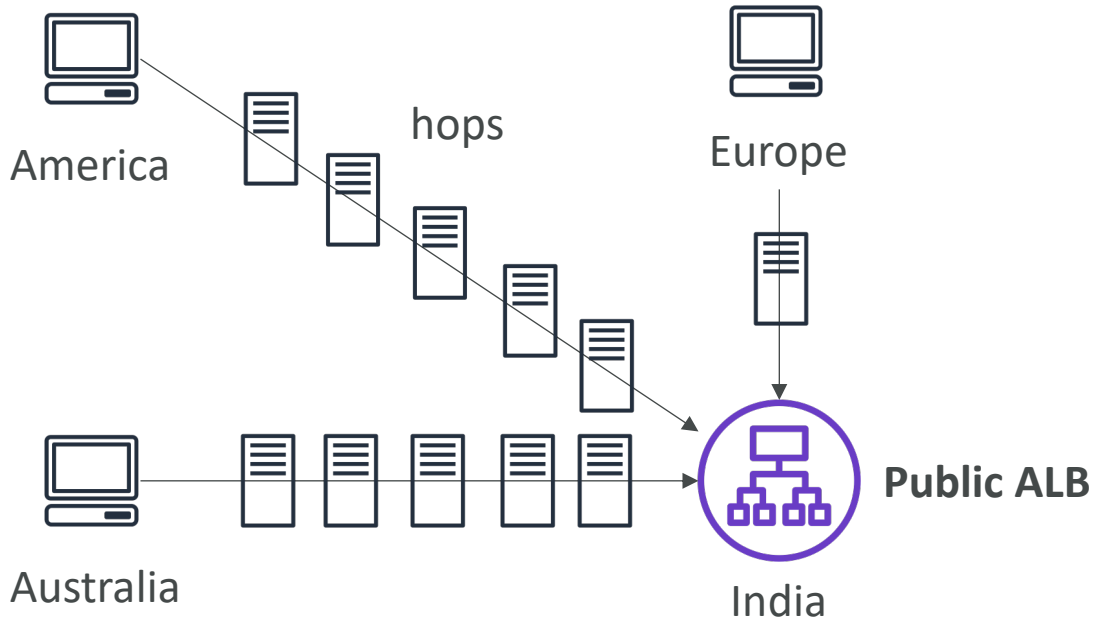


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Global users for our application

- You have deployed an application and have global users who want to access it directly.
- They go over the public internet, which can add a lot of latency due to many hops
- We wish to go as fast as possible through AWS network to minimize latency

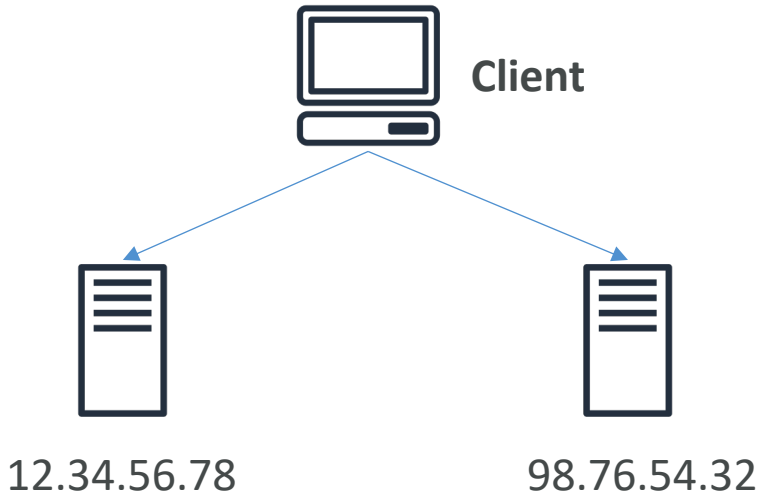


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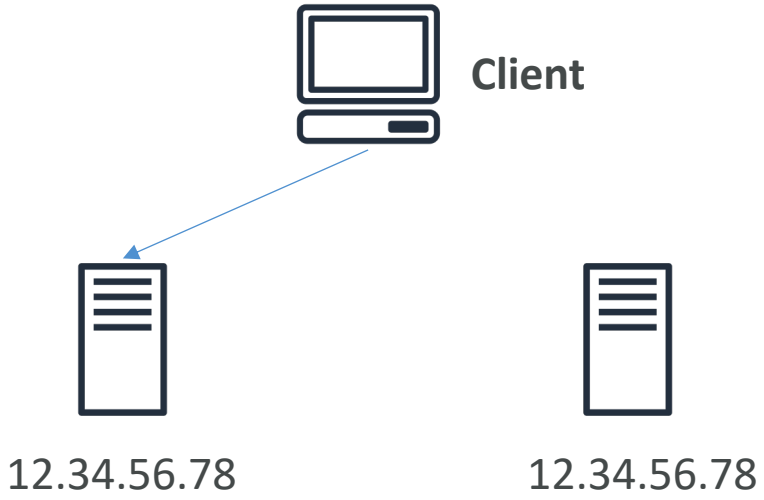
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Unicast IP vs Anycast IP

- **Unicast IP:** one server holds one IP address



- **Anycast IP:** all servers hold the same IP address and the client is routed to the nearest one

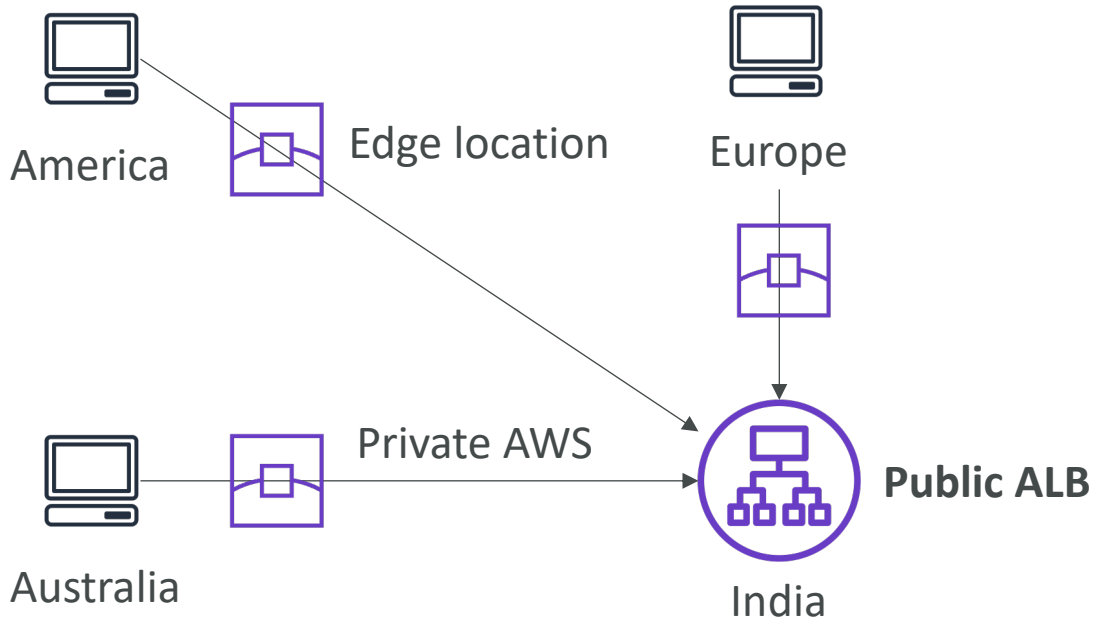


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AWS Global Accelerator

- Leverage the AWS internal network to route to your application
- 2 Anycast IP are created for your application
- The Anycast IP send traffic directly to Edge Locations
- The Edge locations send the traffic to your application



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AWS Global Accelerator

- Works with **Elastic IP, EC2 instances, ALB, NLB, public or private**
- Consistent Performance
  - Intelligent routing to lowest latency and fast regional failover
  - No issue with client cache (because the IP doesn't change)
  - Internal AWS network
- Health Checks
  - Global Accelerator performs a health check of your applications
  - Helps make your application global (failover less than 1 minute for unhealthy)
  - Great for disaster recovery (thanks to the health checks)
- Security
  - only 2 external IP need to be whitelisted
  - DDoS protection thanks to AWS Shield

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AWS Global Accelerator vs CloudFront

- They both use the AWS global network and its edge locations around the world
- Both services integrate with AWS Shield for DDoS protection.

- CloudFront
  - Improves performance for both cacheable content (such as images and videos)
  - Dynamic content (such as API acceleration and dynamic site delivery)
  - Content is served at the edge

- Global Accelerator
  - Improves performance for a wide range of applications over TCP or UDP
  - Proxying packets at the edge to applications running in one or more AWS Regions.
  - Good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP
  - Good for HTTP use cases that require static IP addresses
  - Good for HTTP use cases that required deterministic, fast regional failover

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